



Customer Shopping Behavior Analytics

End-to-end data analyst portfolio project – Python (Pandas) · MySQL · Power BI

PROJECT REPORT

Executive Summary

This project analyzes customer shopping behavior through an end-to-end analytics workflow: Python/Pandas for cleaning, MySQL for SQL analysis, and Power BI for interactive dashboards.

3,900

Customer Records

Unique customer IDs in dataset

\$233K

Total Revenue

\$233,081 recorded purchase revenue

\$59.76

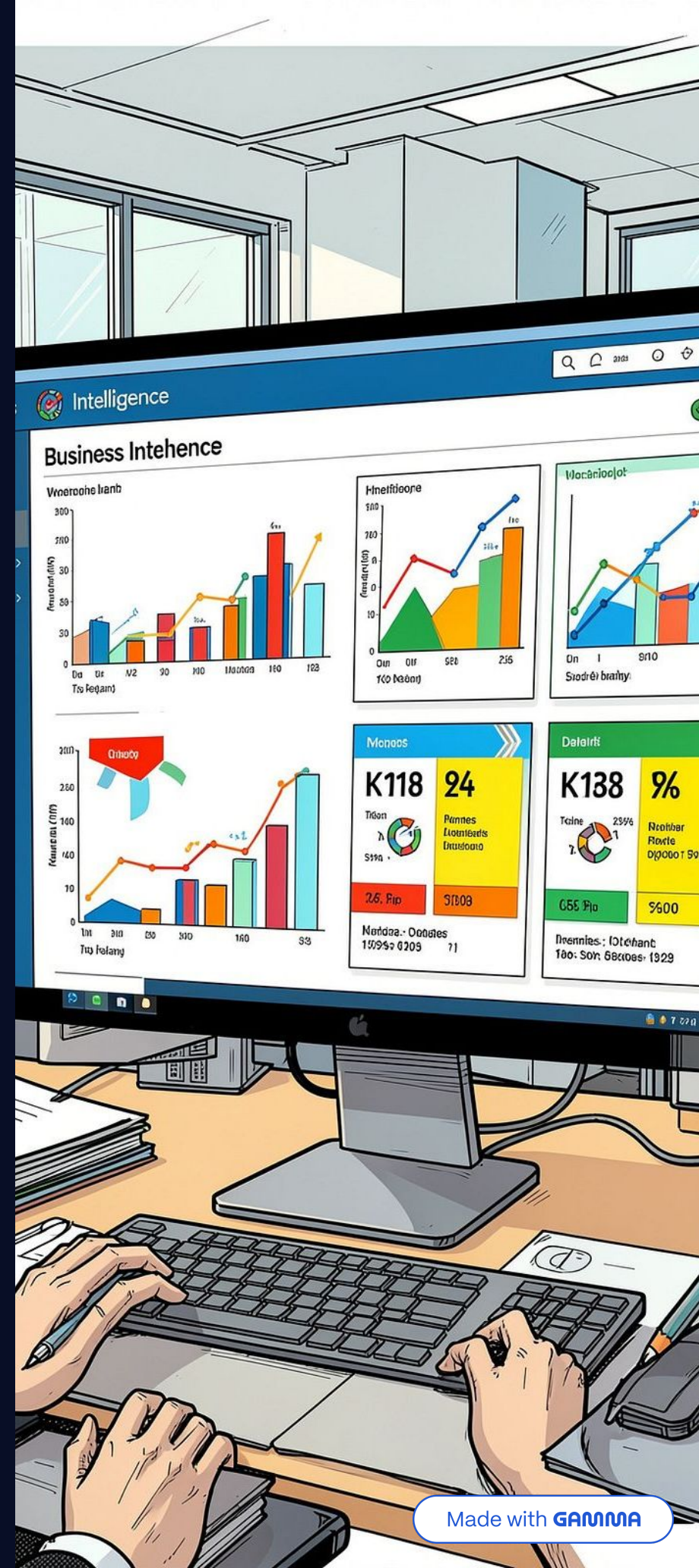
Avg. Purchase

Average purchase amount per record

3.75

Avg. Rating

Average review rating across products



Business Objectives



Purchasing Patterns

Understand customer behavior and revenue contribution across segments.



Subscription Impact

Evaluate whether subscribed customers spend more than non-subscribers.



Customer Segmentation

Segment customers by purchase frequency to support retention strategy.



Product Performance

Identify high-performing products and categories by rating and volume.



Discount & Shipping

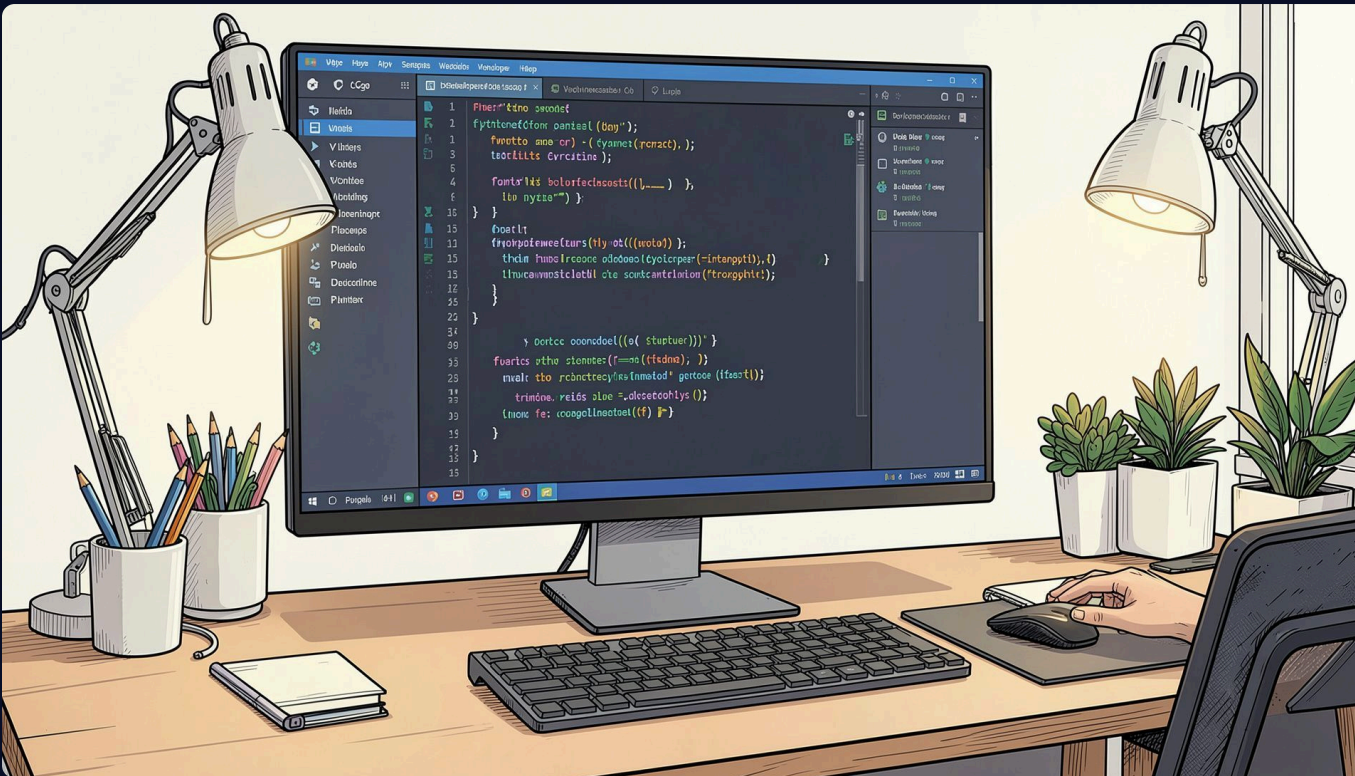
Analyze discount penetration and purchasing behavior across shipping types.



Actionable Insights

Provide recommendations to support promotions, engagement, and product strategy.

Tools & Technology Stack



Python / Pandas

Data cleaning, transformation, and feature engineering

VS Code

Python development environment

MySQL & Workbench

Data storage, SQL analysis, and query execution

Power BI

Interactive dashboarding and visualization

CSV

Raw and cleaned data exchange format

Data Preparation & Transformation

Python/Pandas transformed 3,900 records (18 → 19 columns) into a clean dataset.

01

Missing Ratings

Imputed with category-level median review rating.

02

Column Standardization

Lowercase names; renamed `purchase_amount_(usd)` to `purchase_amount`.

03

Age Grouping

Created `age_group` via quartile-based `qcut`: Young Adult, Adult, Middle Age, Senior.

04

Frequency Mapping

Mapped frequency labels to numeric days (7, 14, 30, 90, 365).

05

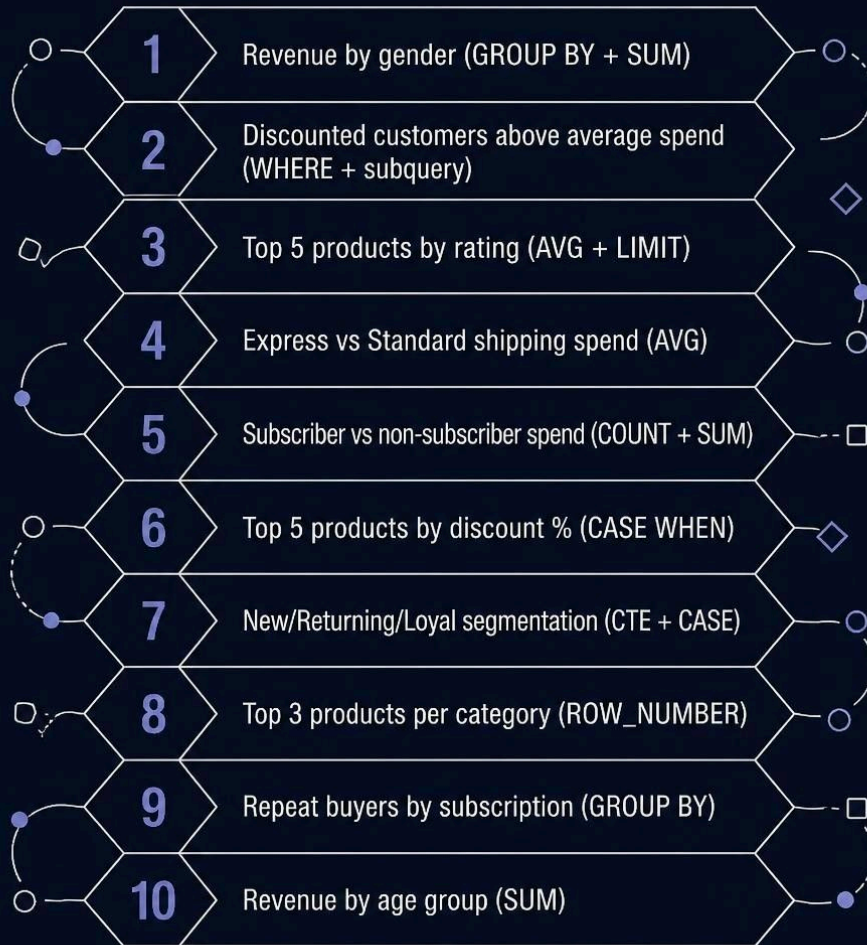
Redundant Field Removal

Removed `promo_code_used` (identical to `discount_applied`).

📄 Output saved as `clean_data.csv` with 19 fields.

SQL Business Analysis

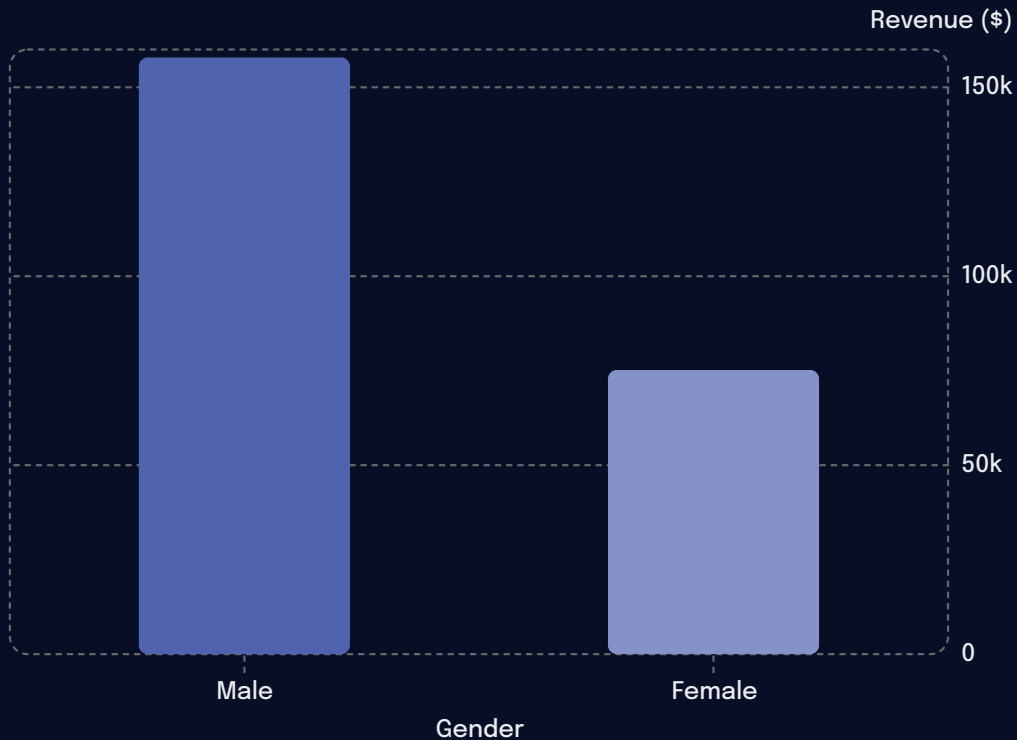
10 business questions answered using aggregation, subqueries, CTEs, and window functions.



SQL Techniques Used

- GROUP BY, SUM, AVG, COUNT
- Subqueries and WHERE filters
- CTEs for complex segmentation
- ROW_NUMBER() window function
- CASE WHEN for conditional logic

Key Findings: Revenue & Subscription



Revenue by Gender

Male customers generated **\$157,890** vs. **\$75,191** from females – driven by record composition, not causal preference.

Subscription Behavior

Non-subscribers (2,847 customers) generated **\$170,436** vs. subscribers (1,053) at **\$62,645**. Average spend is nearly identical (~\$59.70), so subscription status alone shows no meaningful spend advantage.

Key Findings: Products, Shipping & Segments

Top-Rated Products

Product	Avg. Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Handbag	3.78

Highest Discount Penetration

Product	Discounted %
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Express Shipping

Avg. purchase **\$60.48** vs. Standard **\$58.46**

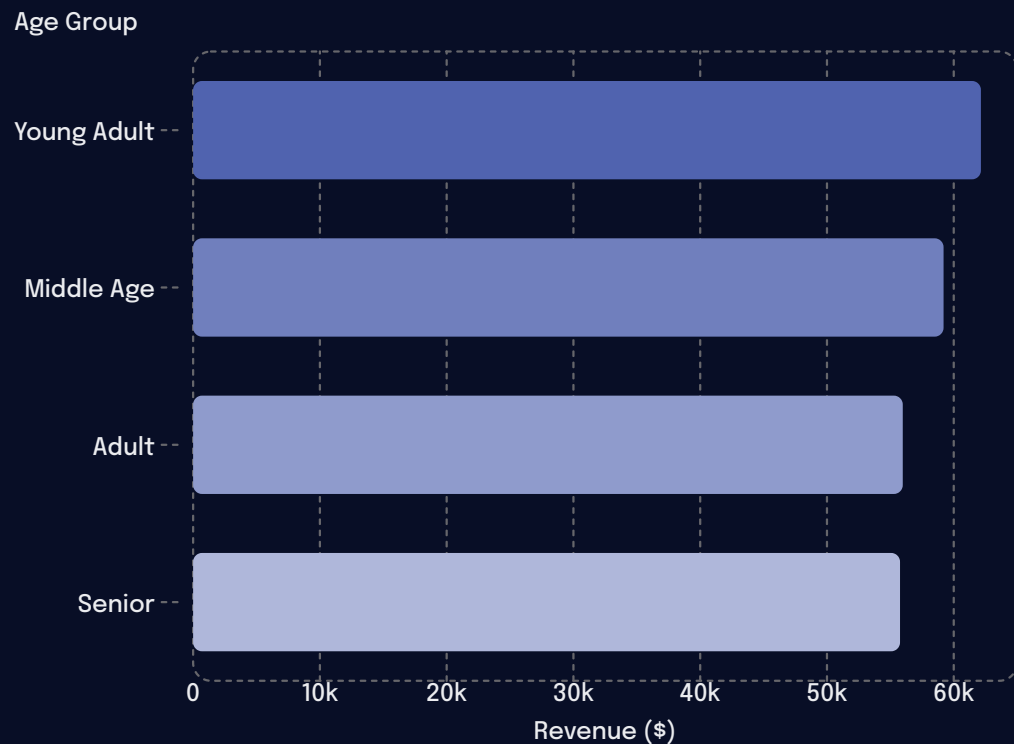
Customer Segments

Loyal: 3,116 · Returning: 701 · New: 83

Repeat Buyers (5+ purchases)

Non-subscribers: 2,518 ·
Subscribers: 958

Revenue by Age Group & Power BI Dashboard



Power BI Dashboard

The cleaned data was loaded into Power BI to build an interactive Customer Behavior Dashboard with:

- KPI cards for customer count, revenue, avg. purchase, and ratings
- Category and product visuals for high-performing items
- Customer and subscription comparison charts
- Demographic and geographic views
- Filters/slicers for customer, product, season, category, and subscription segments

Recommendations & Limitations

Business Recommendations

- Test targeted subscription incentives – subscribers don't inherently spend more.
- Review high-discount products to assess if discounts drive incremental demand.
- Use top-rated products in promotions; monitor Express shipping for larger baskets.
- Prioritize Returning → Loyal retention; tailor messaging by age group.

Limitations & Future Work

- Static snapshot – no time-series trend analysis
- No cost/profit fields – margin analysis not possible
- Unique customer IDs limit longitudinal analysis
- Future: add order dates, cost, returns, cohorts, RFM, CLV

📄 **Workflow:** CSV → Pandas → MySQL → Power BI → Insights